

UC Berkeley Math 128A: Homework 4

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MATLAB usage classification: PEN (pen and paper only), CALC (use as calculator), CODE (run codes).

0. Please assign yourself a self grade on this homework (0, 1, or 2) based off the guidelines in the syllabus.
1. (3.1-5a, CALC) Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:
 - a. $f(8.4)$ if $f(8.1) = 16.9440989$, $f(8.3) = 17.5649208$, $f(8.6) = 18.5051549$, $f(8.7) = 18.8209103$
2. (3.1-7a, CALC) The data in the previous exercise were generated using the following function. Use the error formula to find a bound for the error, and compare the bound to the actual error for the cases $n = 1$ and $n = 2$.
 - a. $f(x) = x \ln x$
3. (3.3-3a, CALC) Use the Newton forward-difference formula to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.
 - a. $f(-\frac{1}{3})$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$
4. (3.3-5a, CALC) Repeat the previous exercise using the Newton backward-difference formula.
5. (3.3-16, PEN) For a function f , the forward-divided differences are given by

$$\begin{array}{llll} x_0 = 0.0 & f[x_0] & & \\ & & f[x_0, x_1] & \\ x_1 = 0.4 & f[x_1] & & f[x_0, x_1, x_2] = \frac{50}{7} \\ & & f[x_1, x_2] = 10 & \\ x_2 = 0.7 & f[x_2] = 6 & & \end{array}$$

Determine the missing entries in the table.

6. (3.4-2a, CALC) Use the Hermite theorem or divided differences to construct an approximating polynomial for the following data.
 - a.

x	$f(x)$	$f'(x)$
0	1.00000	2.00000
0.5	2.71828	5.43656

7. (3.4-4a, CALC) The data in the previous exercise was generated using the following function. Use the polynomials constructed for the given value of x to approximate $f(x)$, and calculate the absolute error.
 - a. $f(x) = e^{2x}$; approximate $f(0.43)$.
8. (3.4-7/9, CALC) The following table lists data for the function described by $f(x) = e^{0.1x^2}$. Approximate $f(1.25)$ by using $H_5(1.25)$ and $H_3(1.25)$, where H_5 uses the nodes $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$; and H_3 uses the nodes $\bar{x}_0 = 1$ and $\bar{x}_1 = 1.5$. Find error bounds for these approximations.

x	$f(x) = e^{0.1x^2}$	$f'(x) = 0.2xe^{0.1x^2}$
$x_0 = \bar{x}_0 = 1$	1.105170918	0.2210341836
$\bar{x}_1 = 1.5$	1.252322716	0.3756968148
$x_1 = 2$	1.491824698	0.5967298792
$x_2 = 3$	2.459603111	1.475761867

9. (3.4-11a, PEN) Show that $H_{2n+1}(x)$ is the unique polynomial of least degree agreeing with f and f' at $x_0, \dots, x_n$. [Hint : Assume that $P(x)$ is another such polynomial and consider $D = H_{2n+1} - P$ and D' at $x_0, x_1, \dots, x_n$.]